

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

The database server is critical to business operations because it stores sensitive customer information, financial records, and internal business data. This data supports daily operations, reporting, and decision-making processes.

Securing the server is essential to:

- Protect sensitive customer and financial data
- Maintain regulatory compliance
- Preserve business reputation
- Ensure data integrity and availability

If the server were disabled, the organization could experience:

- Operational downtime
- Financial losses
- Data loss or corruption
- Reputational damage
- Potential legal or compliance penalties

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>External attacker</i>	<i>Brute-force attack on DB credentials</i>	2	3	3
<i>Malicious insider</i>	<i>Unauthorized data exfiltration</i>	2	3	3
<i>System</i>	<i>System failure</i>	1	3	3

Approach

The assessment evaluated risks related to database access controls, authentication mechanisms, network exposure, and encryption methods. Likelihood and impact were rated based on the probability of occurrence and the potential business disruption. Risks were prioritized according to their combined severity and likelihood scores to determine appropriate mitigation strategies.

Remediation Strategy

To reduce identified risks, the following controls will be implemented:

Access Control Improvements

- Enforce strong password policies
- Implement multi-factor authentication (MFA)

- Apply role-based access control (RBAC)
- Follow the principle of least privilege
- Conduct periodic access reviews

Network Security Enhancements

- Replace deprecated SSL with TLS 1.2 or higher
- Implement IP allow-listing for corporate locations
- Restrict database exposure from the public internet
- Deploy firewall rules to limit inbound connections

Monitoring and Auditing

- Enable database activity logging
- Implement intrusion detection and prevention systems (IDS/IPS)
- Configure real-time alerts for suspicious login attempts
- Perform regular vulnerability scans

Availability Protections

- Implement regular automated backups
- Test disaster recovery procedures
- Configure redundancy or failover systems